Week 2, July-Sept 2017
MOOCS Decision Modeling Course

Decision Analysis

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Bayes Theorem

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Estimating Probability by Bayesian Analysis

- Management experience or intuition
- History
- Existing data
- Need to *revise* probabilities based on new data

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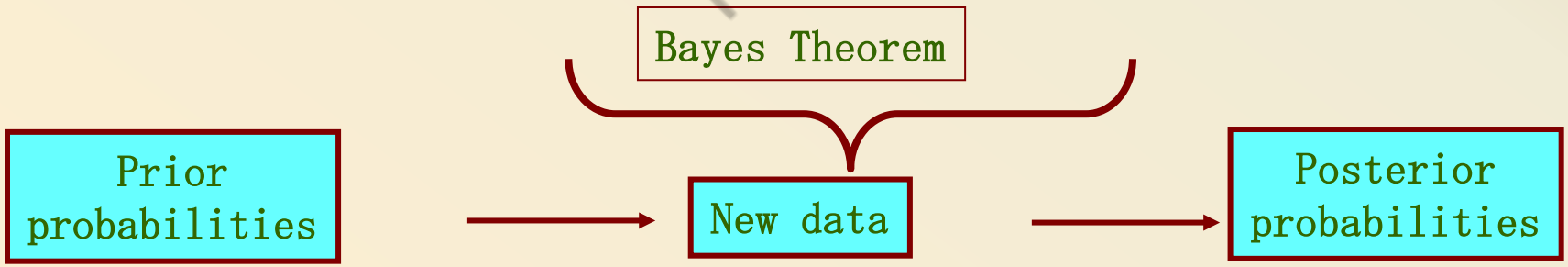
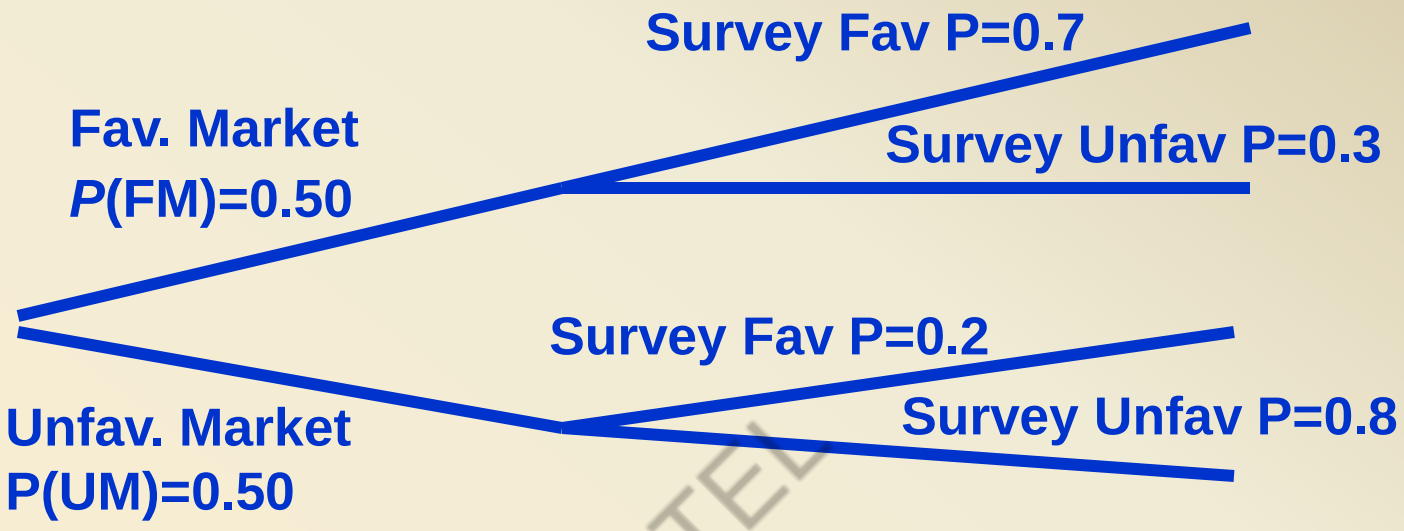


Bayesian Analysis for Probability



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Prior Probabilities: **WITHOUT any market survey information, best estimates of a favorable and unfavorable market are:**

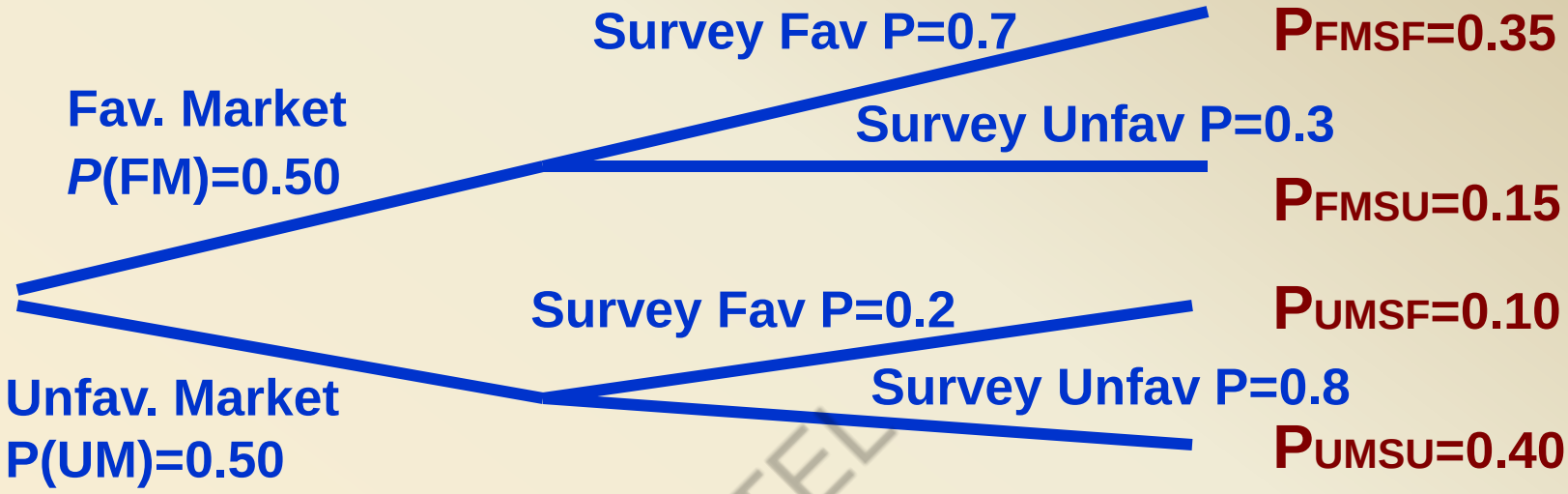
$P(FM) = 0.50$ $\qquad$ $P(UM) = 0.50$

Conditional Probabilities: **WITH market survey, are the following:**

$P(\text{Survey Favorable} \mid FM) = 0.70$ $\qquad$ $P(\text{Survey Favorable} \mid UM) = 0.20$

$P(\text{Survey Unfavorable} \mid FM) = 0.30$ $\qquad$ $P(\text{Survey Unfavorable} \mid UM) = 0.80$

Bayesian Analysis for Probability



Joint Probabilities:

P(Fav. Market U Survey Fav.) = 0.5x0.7 = 0.35

P(Fav. Market U Survey Unfav.) = 0.5x0.3 = 0.15

P(Unfav. Market U Survey Fav.) = 0.5x0.2 = 0.10

P(Unfav. Market U Survey Unfav.) = 0.5x0.8 = 0.40

Thus, Total Probabilities:

P(Survey Favorable) = 0.35 + 0.10 = 0.45

P(Survey Unfavorable) = 0.15+0.40 = 0.55

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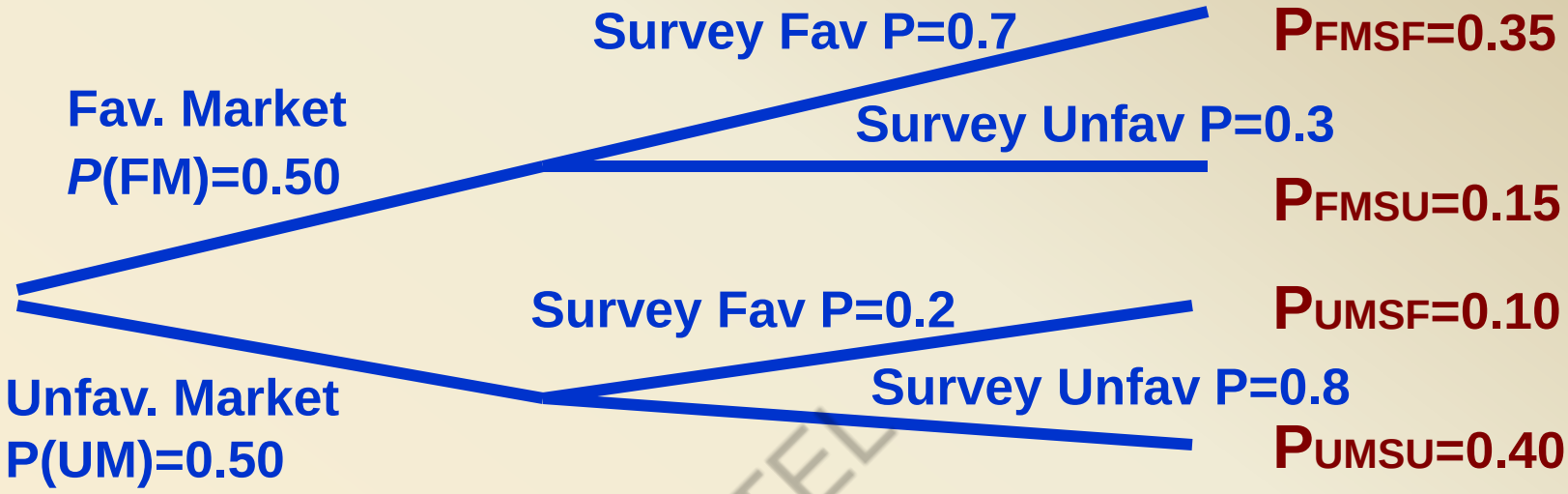
Bayesian Analysis for Probability



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Total Probabilities:

P(Survey Favorable) = 0.45; P(Survey Unfavorable) = 0.55

So, **Posterior Probabilities:**

P(Fav. Market | Survey Favorable) = 0.35/0.45 = 0.78

P(Unfav. Market | Survey Favorable) = 0.10/0.45 = 0.22

P(Fav. Market | Survey Unfavorable) = 0.15/0.55 = 0.27

P(Unfav. Market | Survey Unfavorable) = 0.40/0.55 = 0.73

Estimating Probability by Bayesian Analysis

Given that Survey predicts favorable, Posterior Probability

$$P\left(\text{FM}\middle|\text{Survey Favorable}\right)=\frac{P\left(\text{Survey Favorable}\middle|\text{FM}\right)P\left(\text{FM}\right)}{P\left(\text{Survey Favorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Favorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}$$

$$=\frac{(0.7)(0.5)}{(0.7)(0.5)+(0.2)(0.5)}=\frac{0.35}{0.45}=0.78$$

$$P\left(\text{UM}\middle|\text{Survey Favorable}\right)=\frac{P\left(\text{Survey Favorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}{P\left(\text{Survey Favorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Favorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}$$

$$=\frac{(0.2)(0.5)}{(0.7)(0.5)+(0.2)(0.5)}=\frac{0.10}{0.45}=0.22$$

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$$\textit{Where},\ P\left(\text{Survey Favorable}\right)=P\left(\text{Survey Favorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Favorable}\middle|\text{UM}\right)P\left(\text{UM}\right)$$

Estimating Probability by Bayesian Analysis

Given that Survey predicts unfavorable, Posterior Probability

$$P\left(\text{FM}\middle|\text{Survey Unfavorable}\right)=\frac{P\left(\text{Survey Unfavorable}\middle|\text{FM}\right)P\left(\text{FM}\right)}{P\left(\text{Survey Unfavorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Unfavorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}$$

$$=\frac{(0.3)(0.5)}{(0.3)(0.5)+(0.8)(0.5)}=\frac{0.15}{0.55}=0.27$$

$$P\left(\text{UM}\middle|\text{Survey Unfavorable}\right)=\frac{P\left(\text{Survey Unfavorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}{P\left(\text{Survey Unfavorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Unfavorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}$$

$$=\frac{(0.8)(0.5)}{(0.3)(0.5)+(0.8)(0.5)}=\frac{0.40}{0.55}=0.73$$

$$\textit{Where},\ P\left(\text{Survey Unfavorable}\right)=P\left(\text{Survey Unfavorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Unfavorable}\middle|\text{UM}\right)P\left(\text{UM}\right)$$

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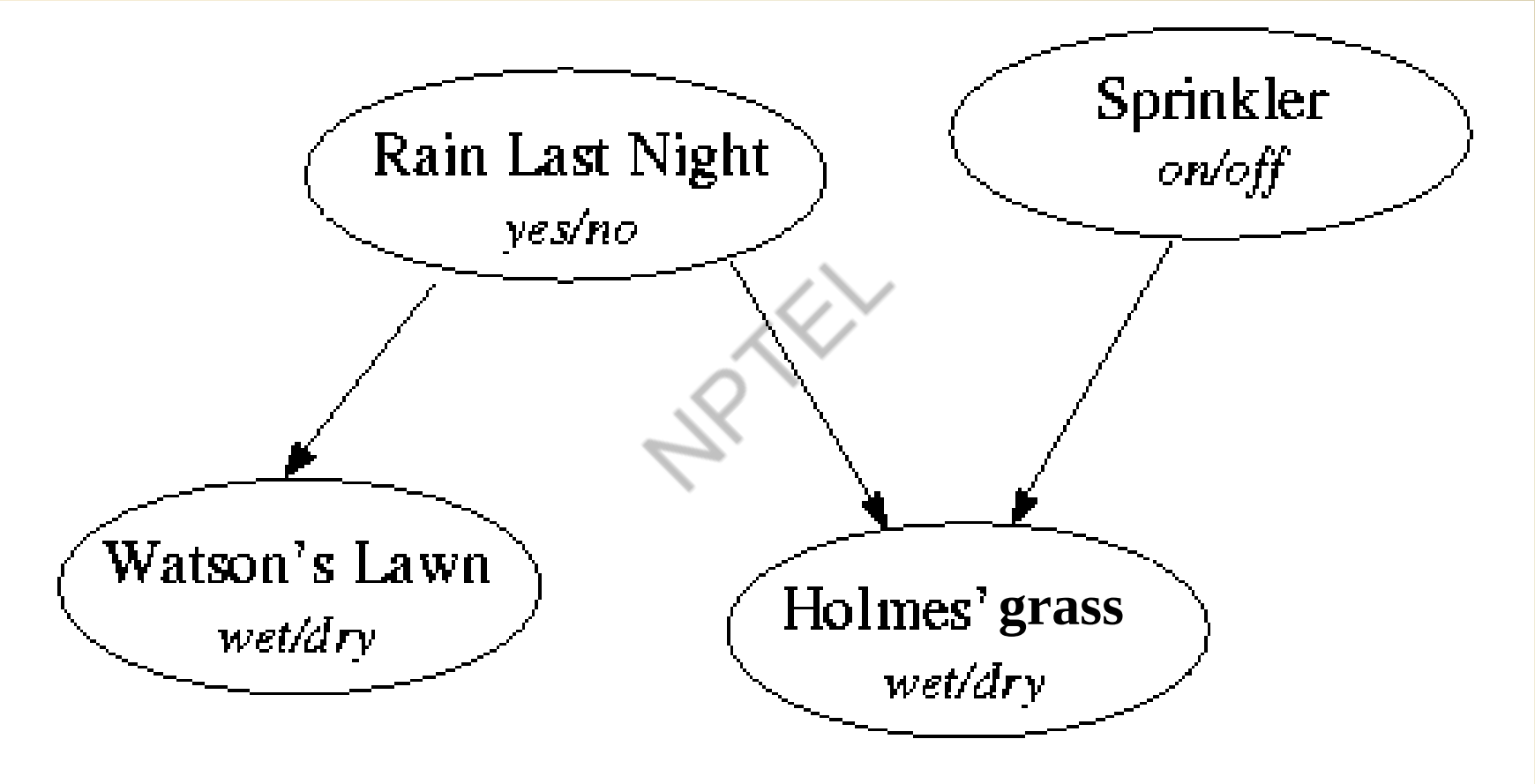
Causal relationships



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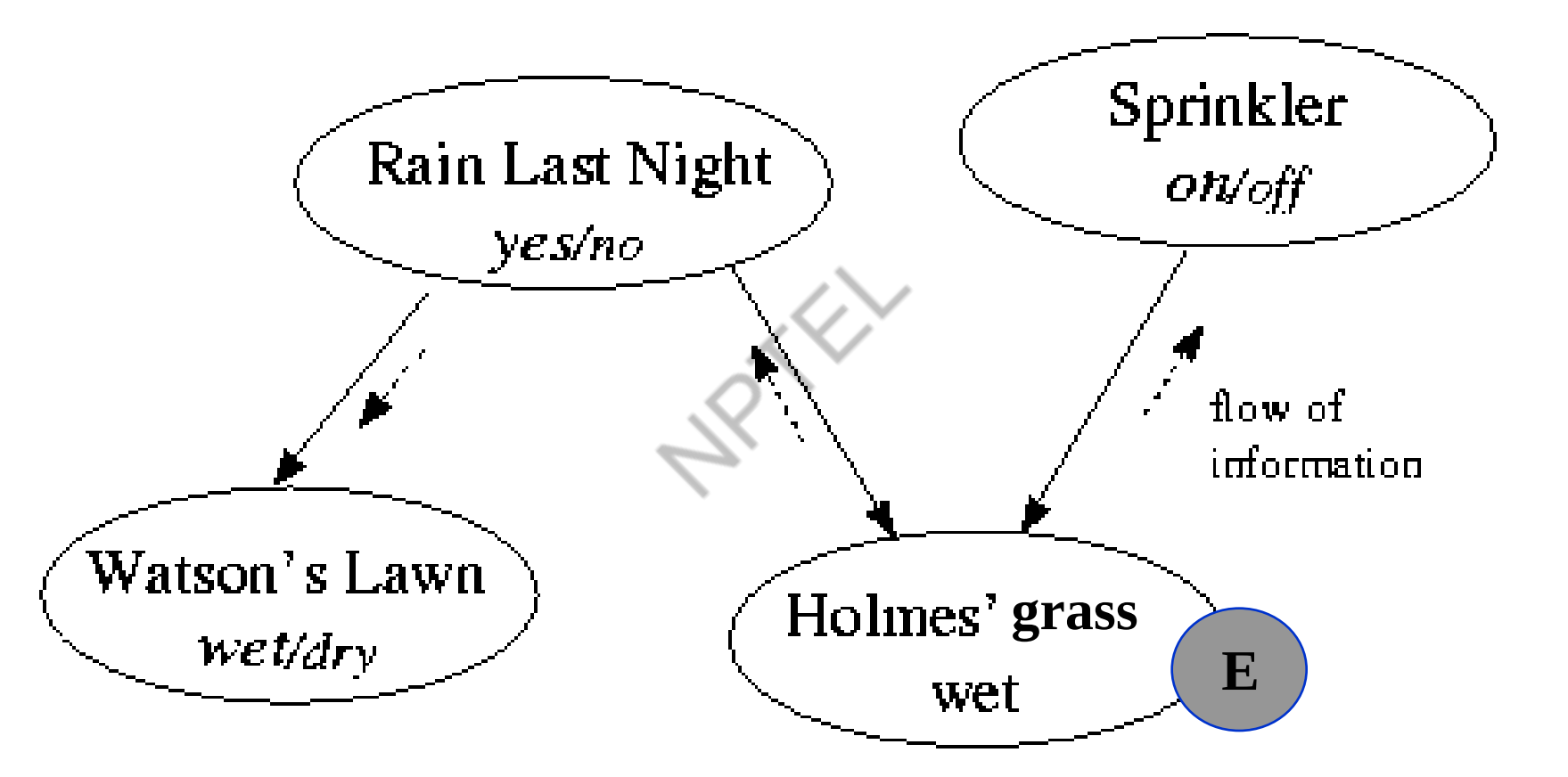
Holmes' grass is wet !



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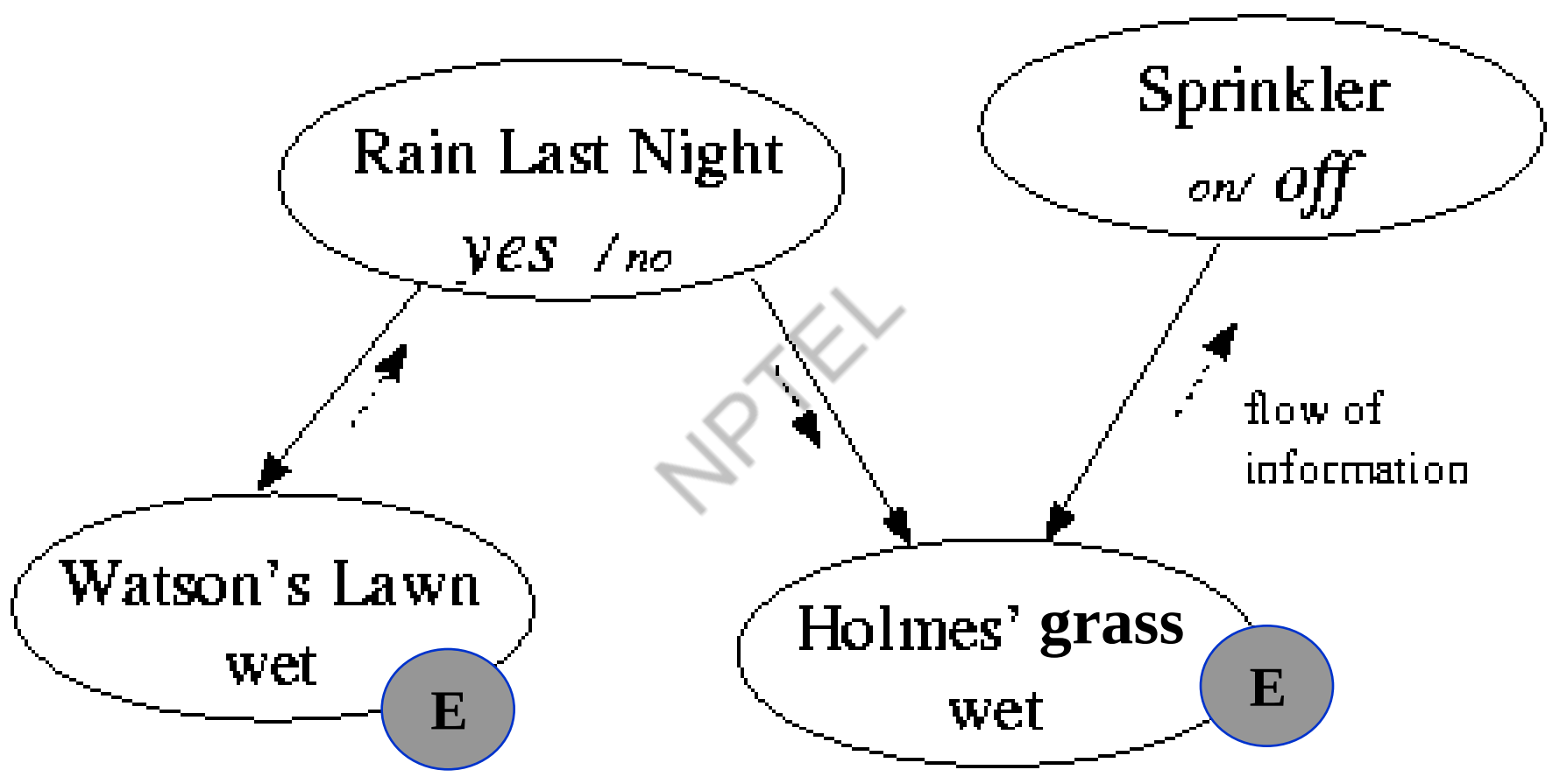
Watson’s lawn is also wet !



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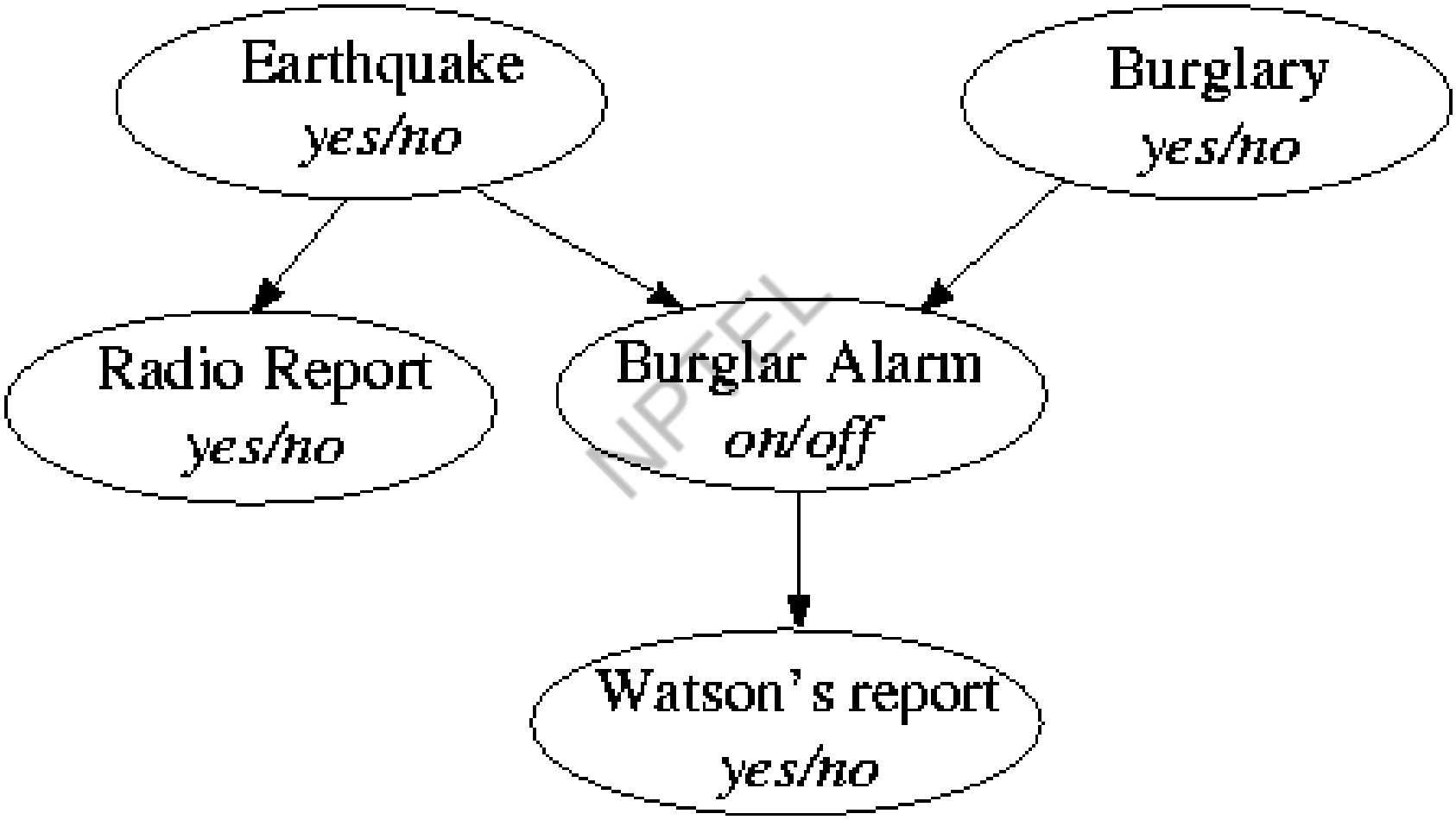
Causal relationships



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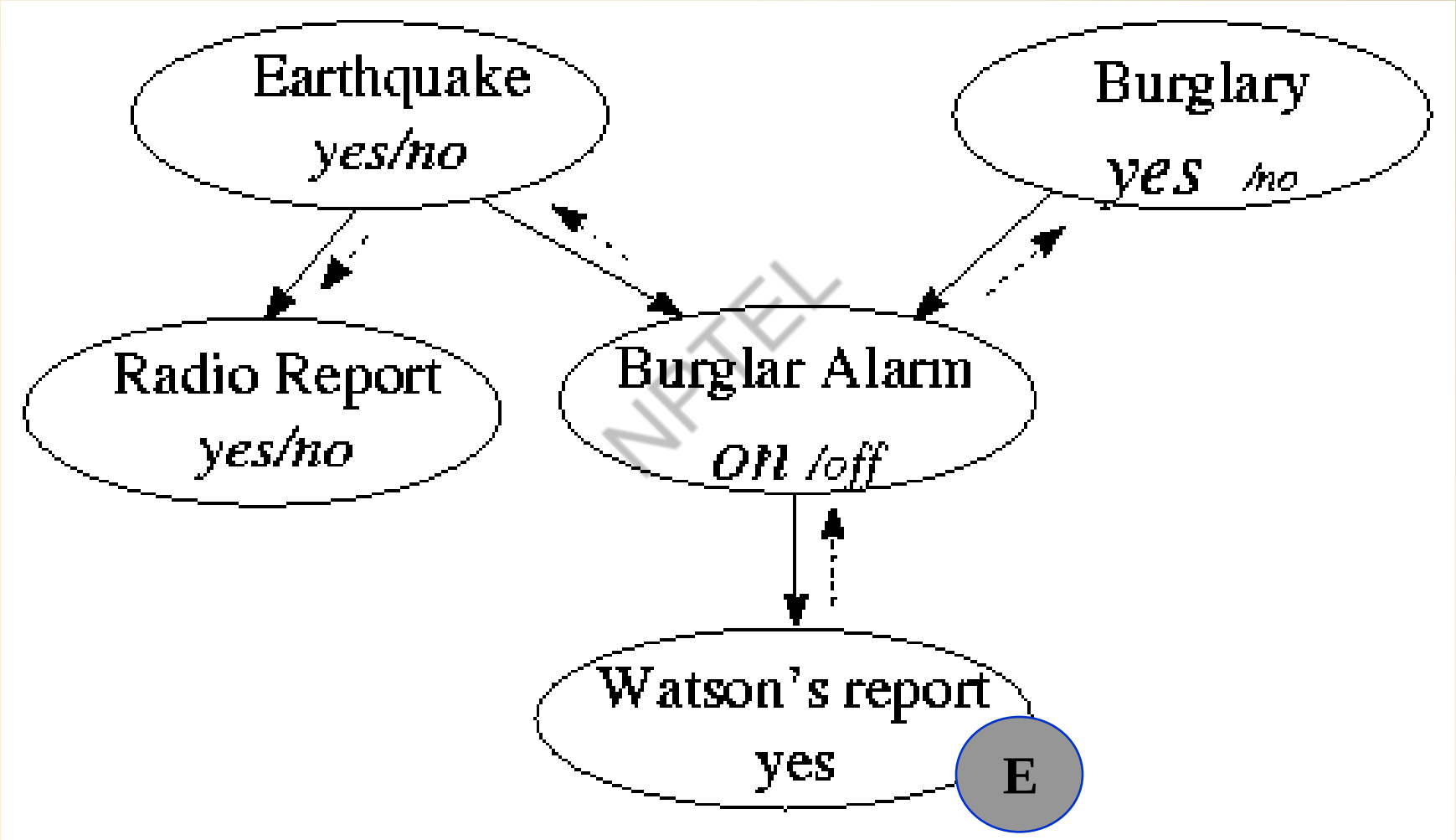
Watson reports about alarm



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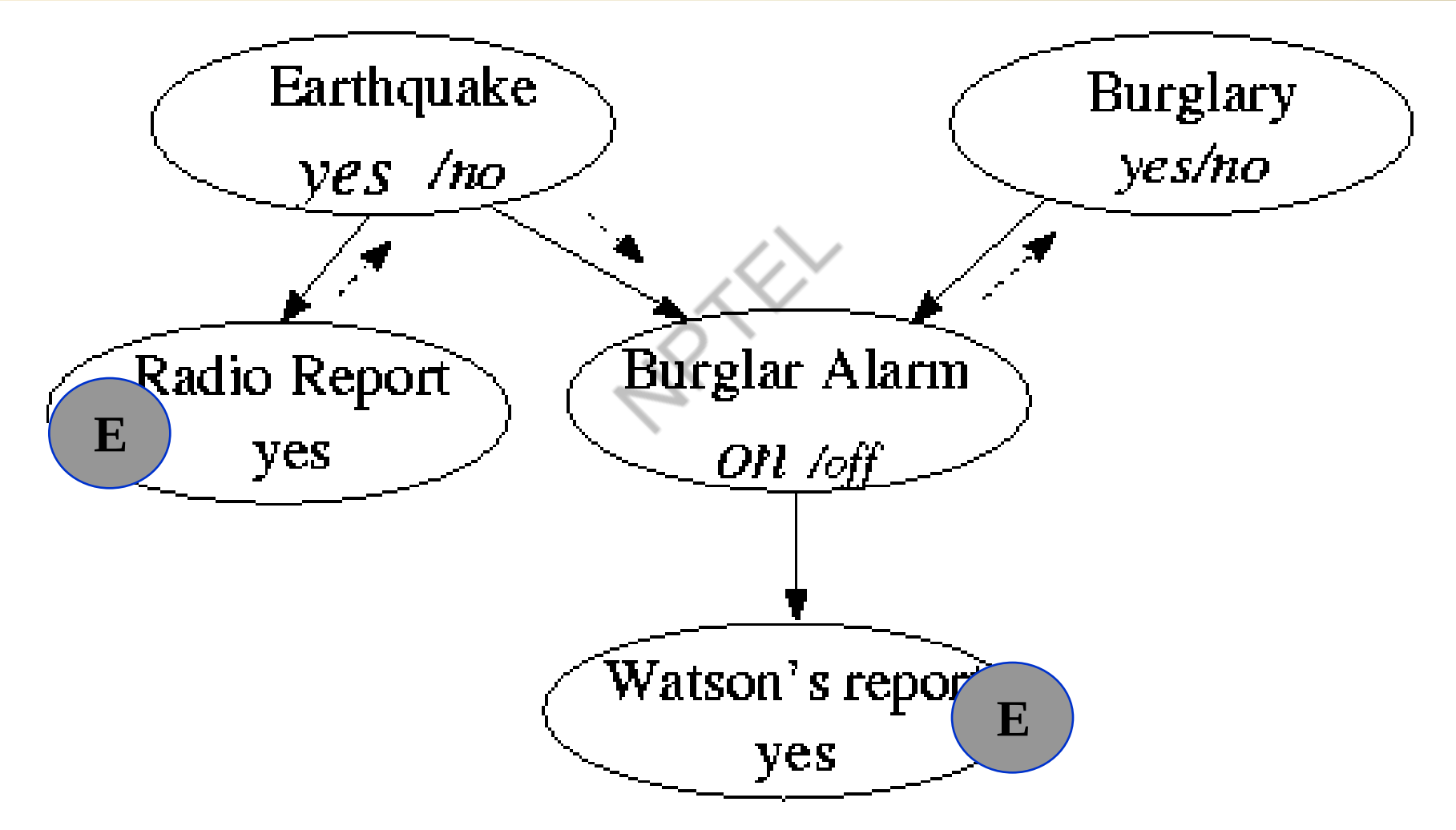
Radio reports about earthquake



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Problem on Bayes Theorem

Suppose there is a certain disease randomly found in one-half of one percent (.005) of the general population. A certain clinical blood test is devised to detect the disease. Clinical tests had found the following conditional probabilities:

	If the disease is present	If the disease is not present
Probability that the test would be positive	0.99	0.05
Probability that the test would be negative	0.01	0.95

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Problem on Bayes Theorem

The disease is present: A;

The disease is not present: ~A

Test result is positive: B;

Test result is negative: ~B

The following are given:

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P(A) = 0.005 **P(~A) = 0.995**

P(B/A) = 0.99 **P(~B/A) = 0.01**

P(B/~A) = 0.05 **P(~B/~A) = 0.95**

P(B) = [P(B/A)*P(A)] + [P(B/~A)*P(~A)]

= 0.99*0.005 + 0.05*0.995 = 0.0547

Problem on Bayes Theorem

Bayes Theorem:

$$P(A/B) = \frac{P(B/A)*P(A)}{(P(B/A)*P(A) + P(B/\sim A)*P(\sim A))}$$

Probability that the disease is present if the test result is positive:

P(A/B) = 0.99*0.005/(0.99*0.005 + 0.05*0.995) = 0.0905

Probability that the disease is not present if the test result is positive (Using Bayes Theorem): P(~A/B)

= P(B/~A)*P(~A)/[P(B/A)*P(A) + P(B/~A)*P(~A)]

= 0.05*0.995/[0.99*0.005 + 0.05*0.995] = 0.9095

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Decision Tree

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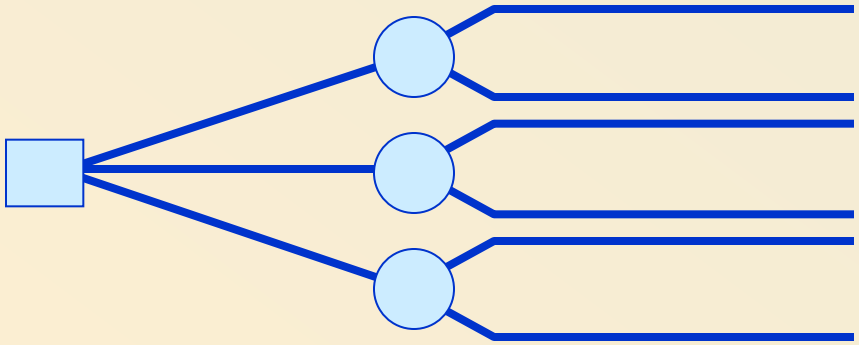
Decision Trees

A **Decision Tree** represents decisions and outcomes in sequential (chronological) order

Symbols

- □ A **decision node** from which one of several alternatives may be selected.
- ○ A **state of nature node** (chance node) out of which one state of nature will occur.
- — An arc (branch) represents possible decisions or states of nature



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Example Problem

ALTERNATIVE	STATE OF NATURE	
	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)
Construct a large plant	200,000	–180,000
Construct a small plant	100,000	–20,000
Do nothing	0	0
Probability	0.5	0.5

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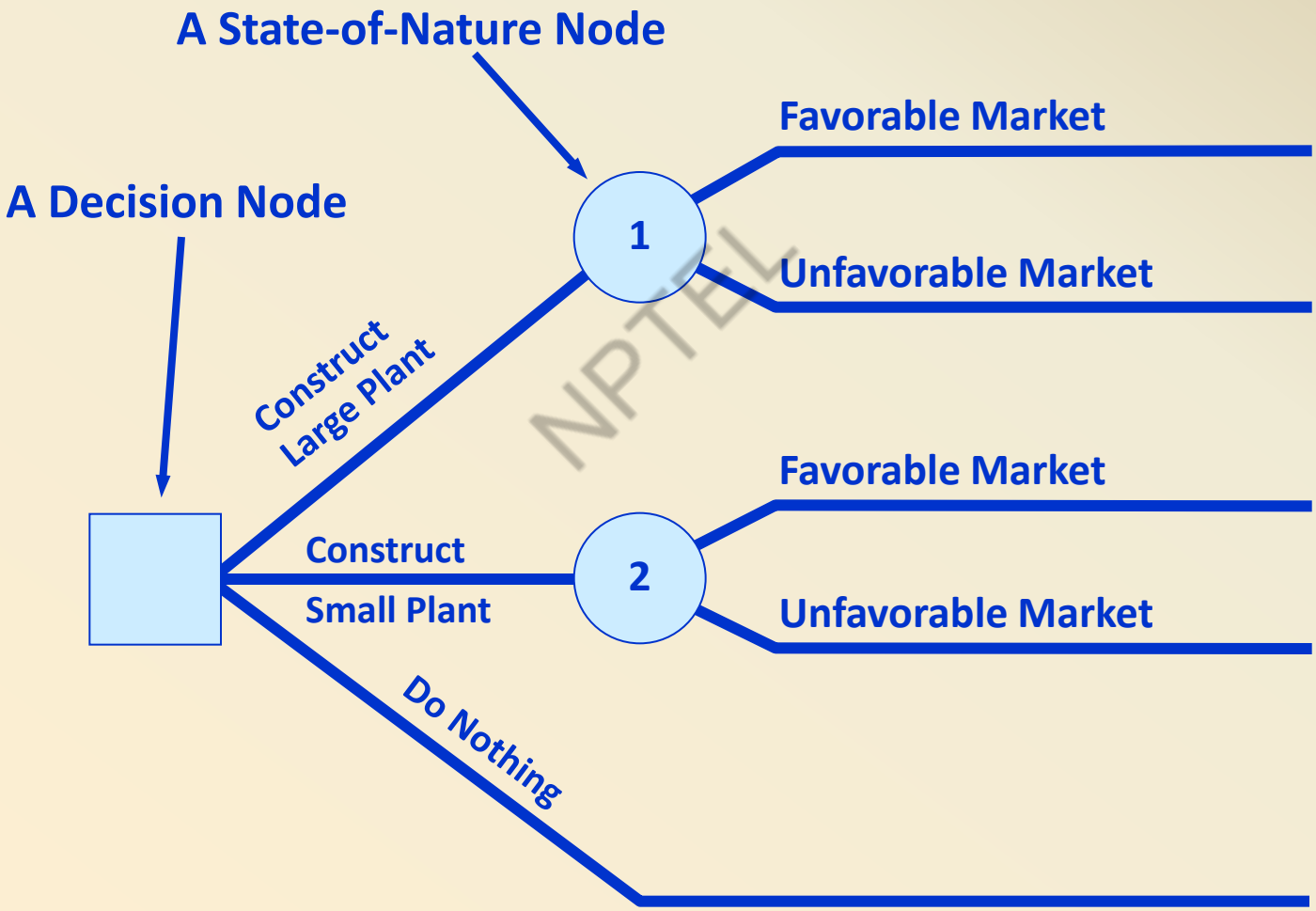
Decision Tree: Example Problem



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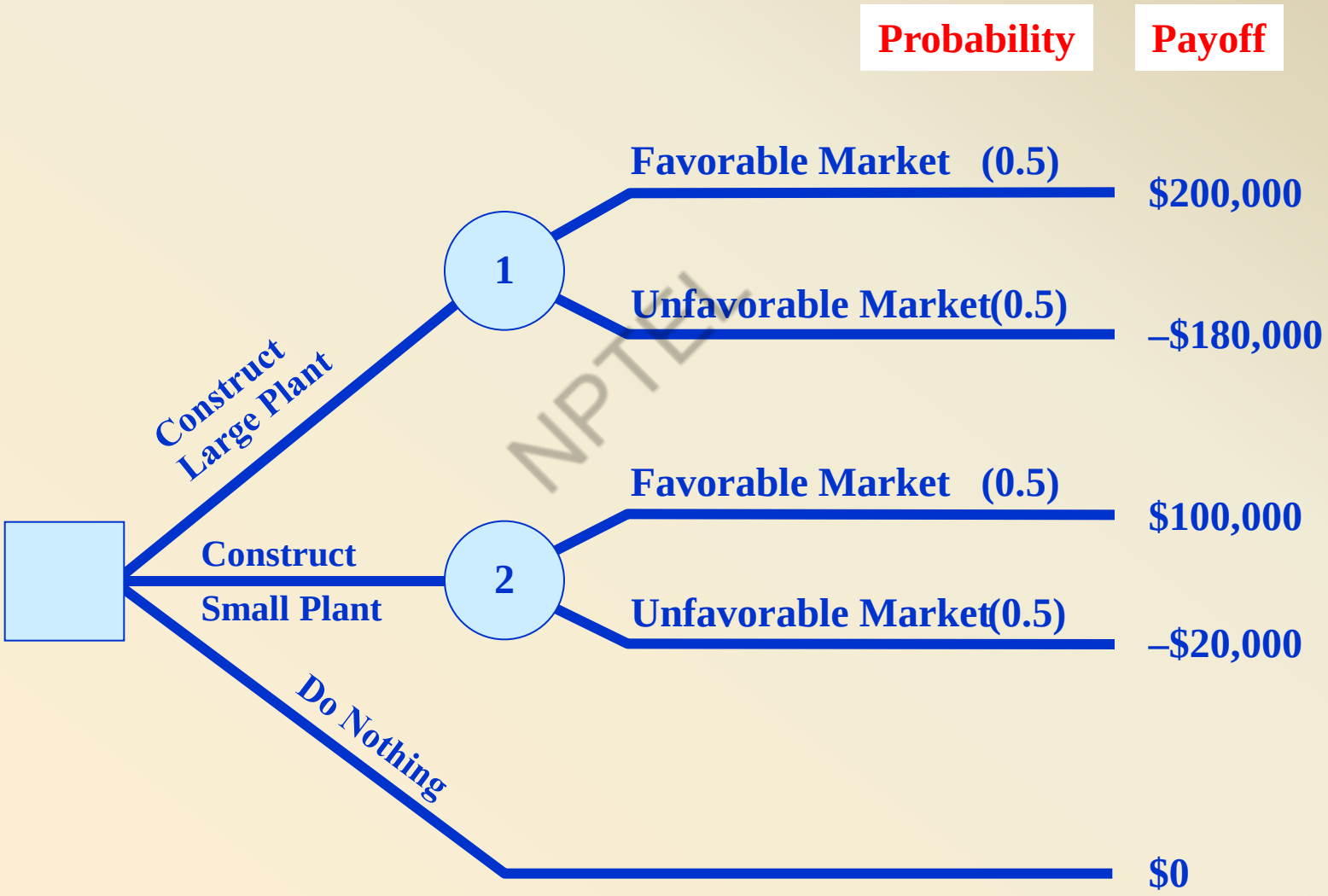
Decision Tree: Example Problem



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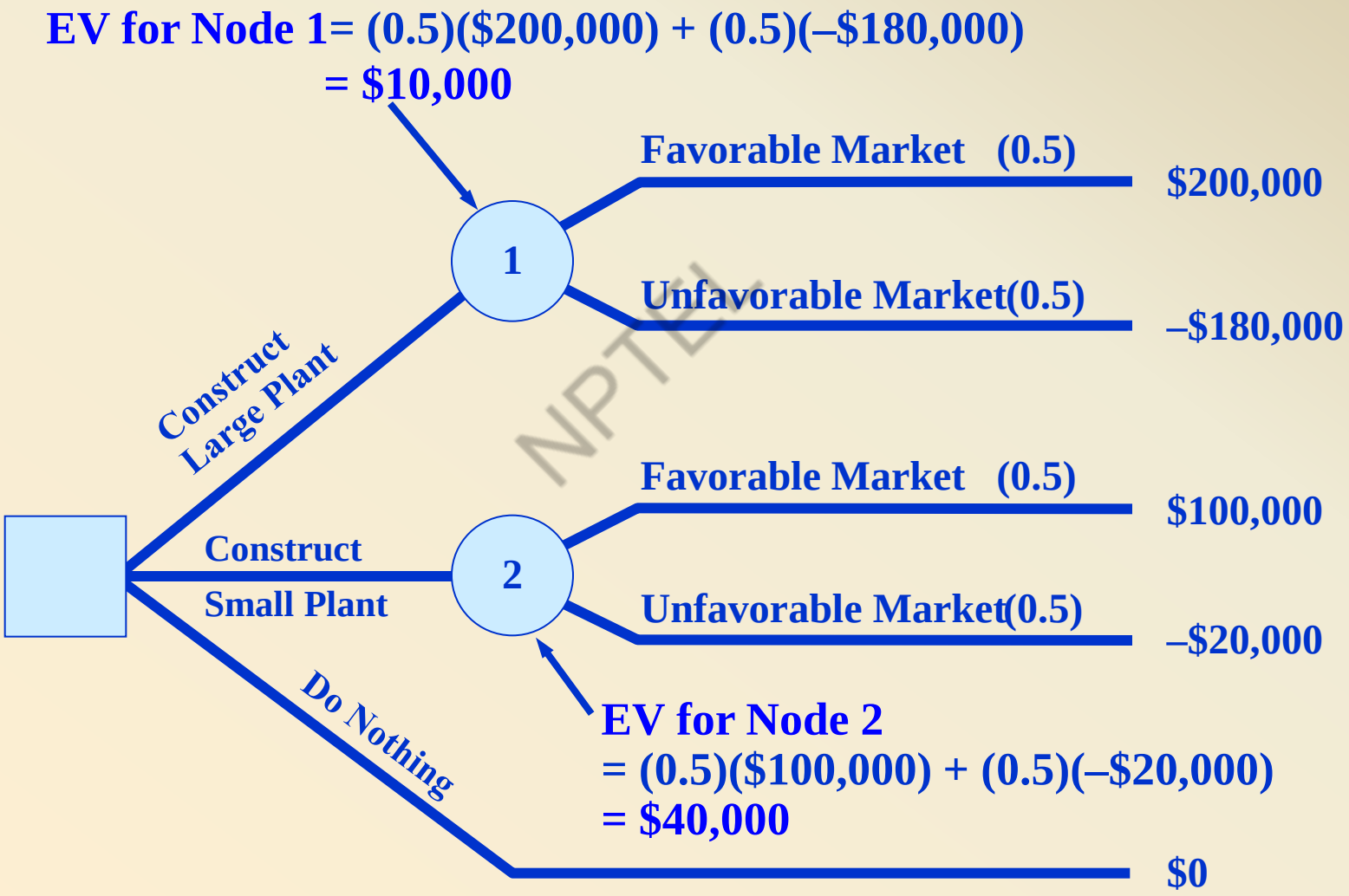
Decision Tree: Example Problem



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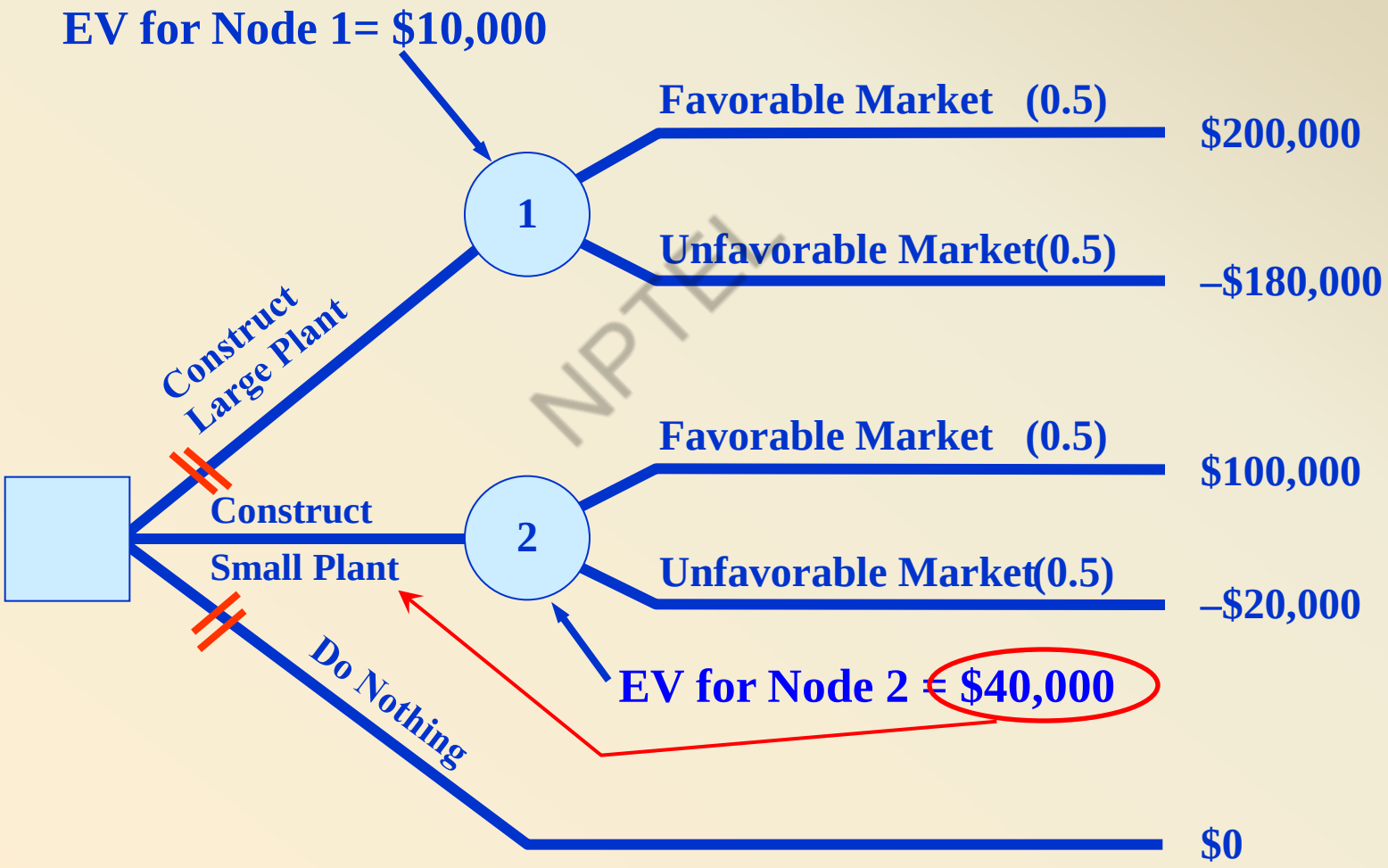
Decision Tree: Example Problem



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Decision Tree and Payoff Matrix

	STATE OF NATURE		
Decision Alternatives	No Market Changes	Favorable market changes	Unfavorable Market changes
Redesign on a small scale	30	100	–80
Rebuild and Refurbish	50	200	–200
Do nothing	0	30	–50

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How do we convert this payoff matrix to a decision tree?

Decision Tree and Payoff Matrix

	STATE OF NATURE		
Decision Alternatives	No Market Changes	Favorable market changes	Unfavorable Market changes
Redesign on a small scale	30	100	–80
Rebuild and Refurbish	50	200	–200
Do nothing	0	30	–50

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How do we convert this payoff matrix to a decision tree?

Example Problem 2

ALTERNATIVE	STATE OF NATURE: FAVORABLE MARKET ($)	STATE OF NATURE: UNFAVORABLE MARKET ($)
Construct a large plant	200,000	–180,000
Construct a small plant	100,000	–20,000
Do nothing	0	0
Probability	0.5	0.5

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Consider that the company can conduct a market survey. Based on the result of the market survey, the company can make a decision on plant construction.

Convert this problem to a Decision Tree problem. What would be the payoffs?

Decision Making With Experimentation

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Decision Making with Experimentation

- Consider the Decision Tree Problem on Plant Construction with Market Survey.
- The Market Survey is an experimentation.
- The company now has two decisions to make:

  1) Whether to go for the market survey

  2) If market survey is favorable, then what type of plant to construct Else what type?

- Here the second decision dependent upon the outcome of the first.

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Decision Making with Experimentation

ALTERNATIVE	STATE OF NATURE: FAVORABLE MARKET ($)	STATE OF NATURE: UNFAVORABLE MARKET ($)
Construct a large plant	200,000	–180,000
Construct a small plant	100,000	–20,000
Do nothing	0	0
Probability	0.5	0.5

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- Before deciding about building a new plant, the company has option to conduct its own **marketing survey**, at a cost of **$10,000**.
- Information from survey could help in deciding which **alternative to pursue** (large, small or no plant)

The Experimentation

- **The market survey is the experimentation conducted at a cost of $10,000.**
- **However, the market survey will not be able to predict the state of nature with 100% accuracy.**
- **The market survey can only give additional information on the probable states of nature in the form of the following conditional probabilities:**

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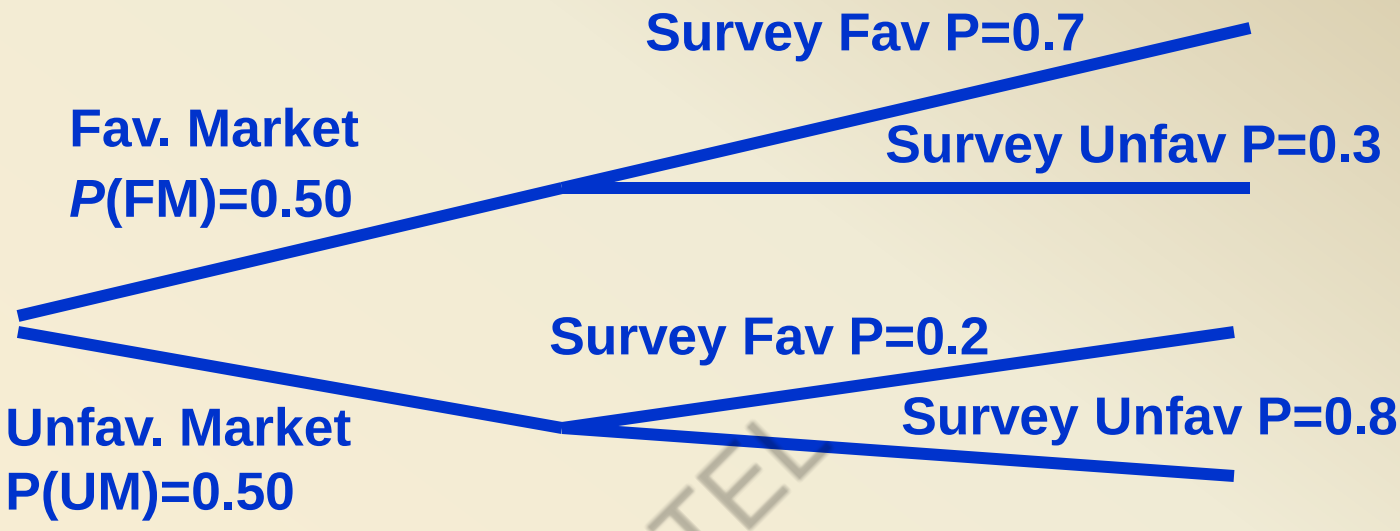
P(Survey Favorable / Favorable Market) = 0.70

P(Survey Unfavorable / Favorable Market) = 0.30

P(Survey Favorable / Unfavorable Market) = 0.20

P(Survey Unfavorable / Unfavorable Market) = 0.80

Bayesian Analysis for Probability



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Prior Probabilities: WITHOUT any market survey information, best estimates of a favorable and unfavorable market are:

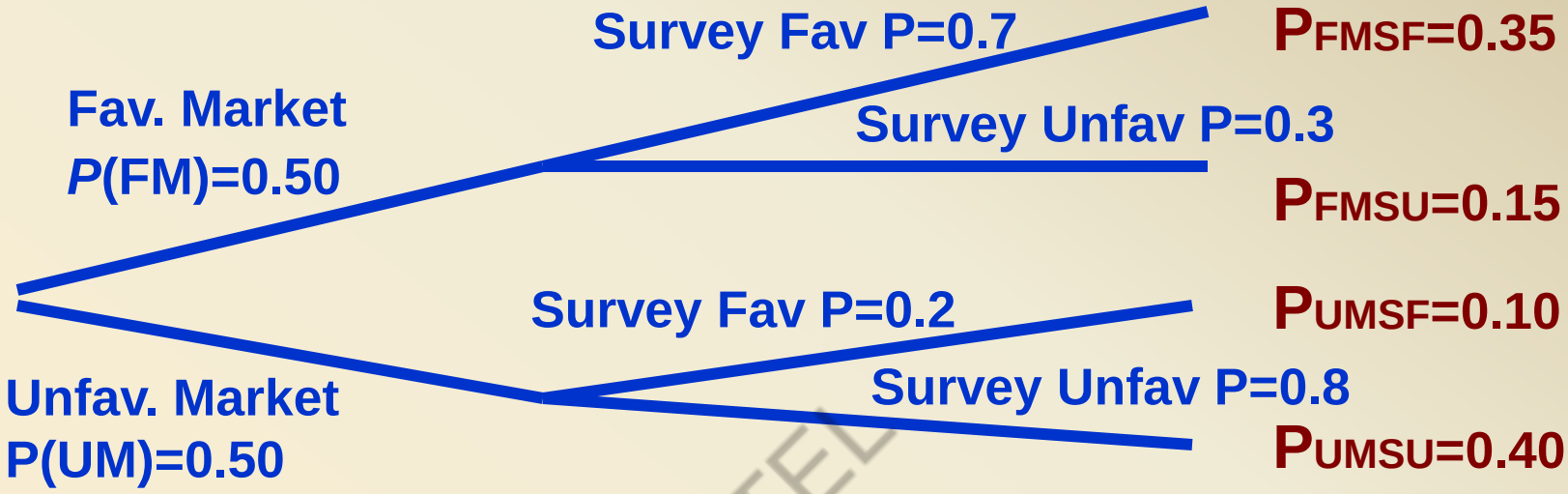
$P(FM) = 0.50$ $\quad$ $P(UM) = 0.50$

Conditional Probabilities: WITH market survey, are the following:

$P(\text{Survey Favorable} / FM) = 0.70$ $\quad$ $P(\text{Survey Favorable} / UM) = 0.20$

$P(\text{Survey Unfavorable} / FM) = 0.30$ $\quad$ $P(\text{Survey Unfavorable} / UM) = 0.80$

Bayesian Analysis for Probability



Joint Probabilities:

P(Fav. Market ∩ Survey Fav.) = 0.5x0.7 = 0.35

P(Fav. Market ∩ Survey Unfav.) = 0.5x0.3 = 0.15

P(Unfav. Market ∩ Survey Fav.) = 0.5x0.2 = 0.10

P(Unfav. Market ∩ Survey Unfav.) = 0.5x0.8 = 0.40

Thus, Unconditional Probabilities:

P(Survey Favorable) = 0.35 + 0.10 = 0.45

P(Survey Unfavorable) = 0.15+0.40 = 0.55

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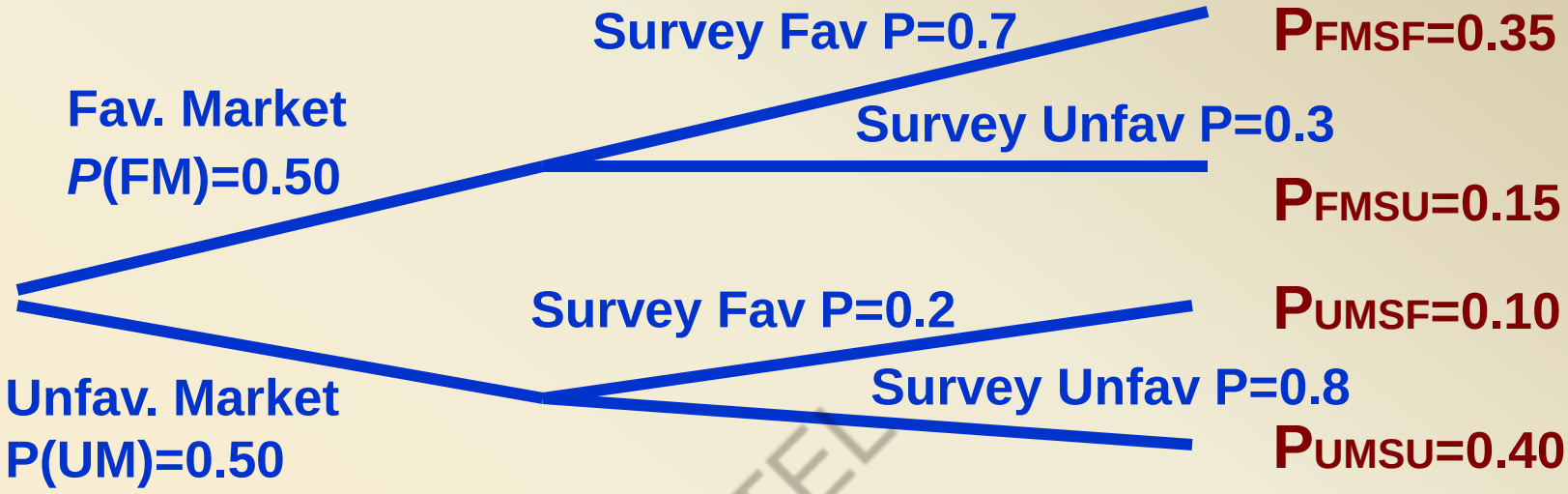
Bayesian Analysis for Probability



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Unconditional Probabilities:

P(Survey Favorable) = 0.45; P(Survey Unfavorable) = 0.55

So, **Posterior Probabilities:**

P(Fav. Market | Survey Favorable) = 0.35/0.45 = 0.78

P(Unfav. Market | Survey Favorable) = 0.10/0.45 = 0.22

P(Fav. Market | Survey Unfavorable) = 0.15/0.55 = 0.27

P(Unfav. Market | Survey Unfavorable) = 0.40/0.55 = 0.73

Estimating Probability by Bayesian Analysis

Given that Survey predicts favorable, Posterior Probability

$$P\left(\text{FM}\middle|\text{Survey Favorable}\right)=\frac{P\left(\text{Survey Favorable}\middle|\text{FM}\right)P\left(\text{FM}\right)}{P\left(\text{Survey Favorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Favorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}$$

$$=\frac{(0.7)(0.5)}{(0.7)(0.5)+(0.2)(0.5)}=\frac{0.35}{0.45}=0.78$$

$$P\left(\text{UM}\middle|\text{Survey Favorable}\right)=\frac{P\left(\text{Survey Favorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}{P\left(\text{Survey Favorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Favorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}$$

$$=\frac{(0.2)(0.5)}{(0.7)(0.5)+(0.2)(0.5)}=\frac{0.10}{0.45}=0.22$$

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$$\textit{Where},\ P\left(\text{Survey Favorable}\right)=P\left(\text{Survey Favorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Favorable}\middle|\text{UM}\right)P\left(\text{UM}\right)$$

Estimating Probability by Bayesian Analysis

Given that Survey predicts unfavorable, Posterior Probability

$$P\left(\text{FM}\middle|\text{Survey Unfavorable}\right)=\frac{P\left(\text{Survey Unfavorable}\middle|\text{FM}\right)P\left(\text{FM}\right)}{P\left(\text{Survey Unfavorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Unfavorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}$$

$$=\frac{(0.3)(0.5)}{(0.3)(0.5)+(0.8)(0.5)}=\frac{0.15}{0.55}=0.27$$

$$P\left(\text{UM}\middle|\text{Survey Unfavorable}\right)=\frac{P\left(\text{Survey Unfavorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}{P\left(\text{Survey Unfavorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Unfavorable}\middle|\text{UM}\right)P\left(\text{UM}\right)}$$

$$=\frac{(0.8)(0.5)}{(0.3)(0.5)+(0.8)(0.5)}=\frac{0.40}{0.55}=0.73$$

Where, $P\left(\text{Survey Unfavorable}\right)=P\left(\text{Survey Unfavorable}\middle|\text{FM}\right)P\left(\text{FM}\right)+P\left(\text{Survey Unfavorable}\middle|\text{UM}\right)P\left(\text{UM}\right)$

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Decision Making With Experimentation (Contd)

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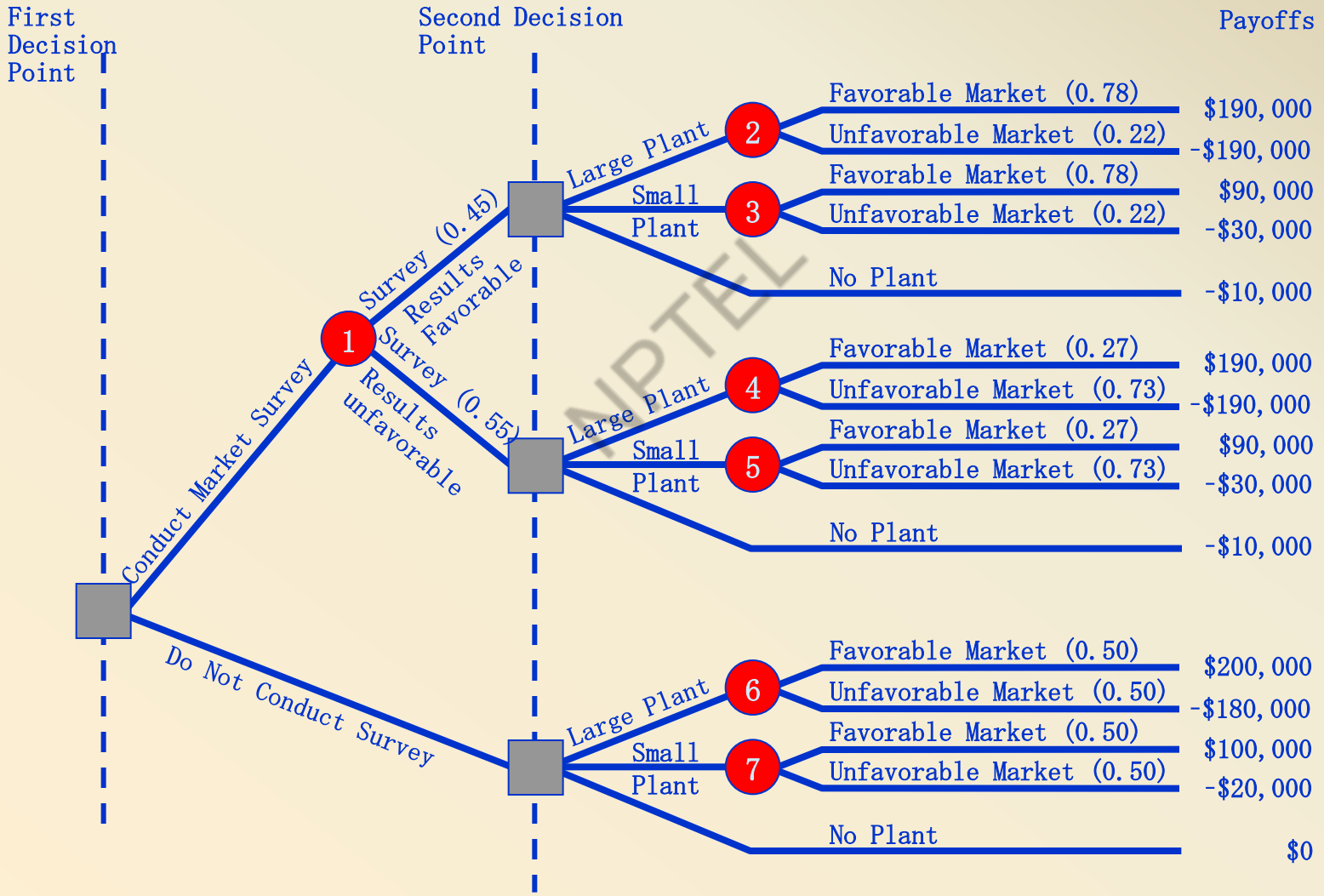
Decision Tree of the Problem

Note: the $10,000 cost was subtracted from each of the first 10 branches. The, $190,000 payoff was originally $200,000 and the $-10,000 was originally $0.



Estimation of the payoffs

1. **Given favorable survey results (positive survey)**

EV(node 2) = EV(large plant | positive survey)

= (0.78)($190,000) + (0.22)(–$190,000) = $106,400

EV(node 3) = EV(small plant | positive survey)

= (0.78)($90,000) + (0.22)(–$30,000) = $63,600

EV for no plant = –$10,000

If survey results are favorable, a large plant should be built with expected value of $106,400

2. **Given unfavorable survey results (negative survey)**

EV(node 4) = EV(large plant | negative survey)

= (0.27)($190,000) + (0.73)(–$190,000) = –$87,400

EV(node 5) = EV(small plant | negative survey)

= (0.27)($90,000) + (0.73)(–$30,000) = $2,400

EV for no plant = –$10,000

If survey results are unfavorable, a small plant should be built with expected value of $2,400

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Estimation of the Payoffs

3. Thus, the expected value of the market survey,

EV(node 1) = EV(conduct survey)

= (0.45)($106,400) + (0.55)($2,400) = $49,200

4. If the market survey is not conducted,

EV(node 6) = EV(large plant)

= (0.50)($200,000) + (0.50)(–$180,000) = $10,000

EV(node 7) = EV(small plant)

= (0.50)($100,000) + (0.50)(–$20,000) = $40,000

EV for no plant = $0

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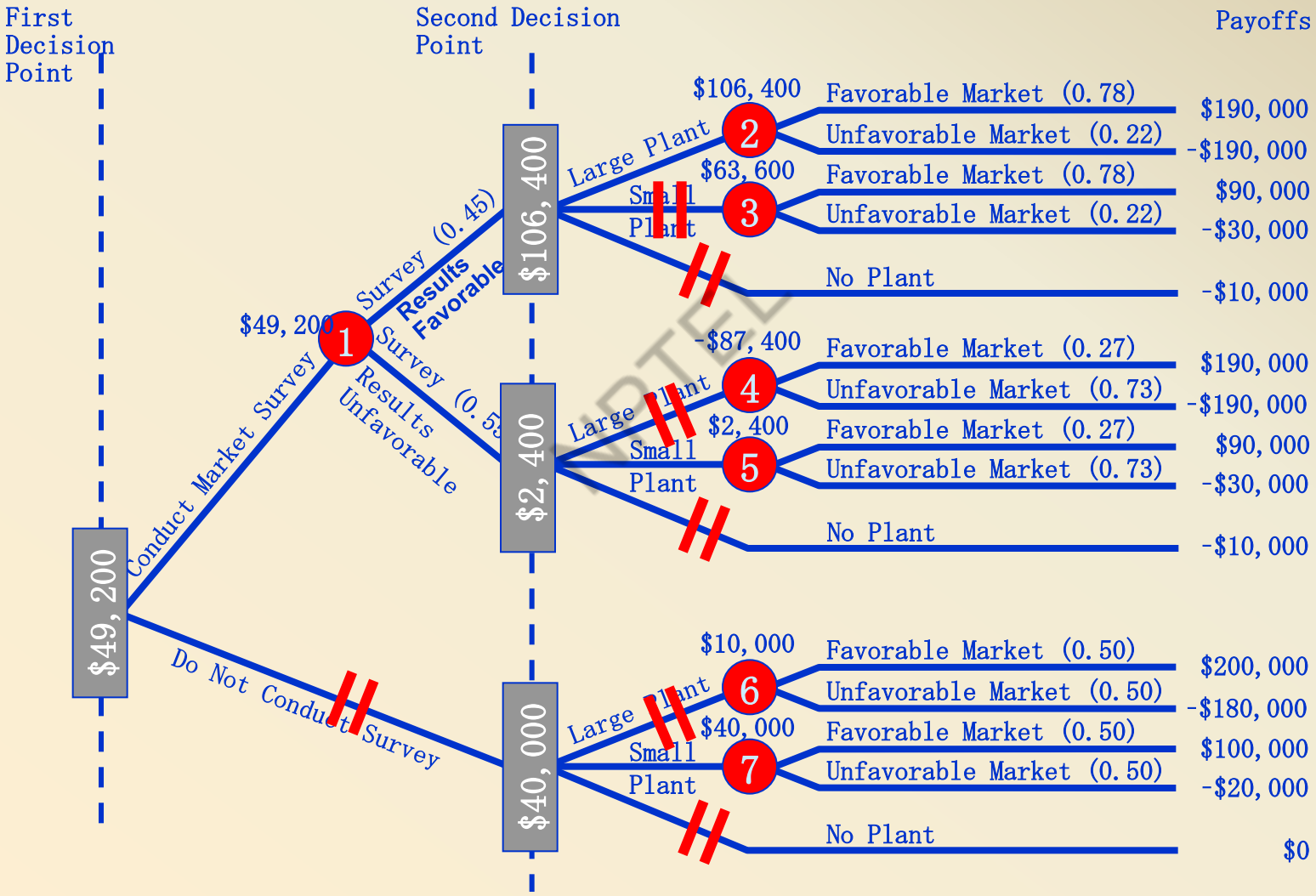
If the market survey is not conducted, a small plant should be built with expected value of $40,000

Decision: Best choice is to seek marketing information.

If survey results are favorable, a large plant should be built or

If survey results are unfavorable, a small plant should be built

Showing Expected Values on Decision Tree



Company's Dilemma

Company is not sure how much value to place on market survey. They want to determine the monetary worth of the survey. Company is also interested in how sensitive the decision is to changes in the market survey results.

- Expected Value of Sample Information (here, market survey)
- Sensitivity Analysis

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Expected Value of Sample Information (EVSI)

- Company want to know the actual value of doing the survey
- Defined as increase in expected value resulting from the sample information

$$\text{EVSI} = \left(\begin{array}{c}\text{Expected value}\\ \text{with sample}\\ \text{information, along with}\\ \text{the cost to gather it}\end{array}\right) - \left(\begin{array}{c}\text{Expected value}\\ \text{of best decision}\\ \text{without sample}\\ \text{information}\end{array}\right)$$

= (EV with sample information + cost) – (EV without sample information)

EVSI = ($49,200 + $10,000) – $40,000 = $19,200

The company could have paid up to $19,200 for a market survey but the survey actually costs $10,000.

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Sensitivity Analysis

How sensitive are the decisions to changes in the probabilities?

- **How sensitive is our decision to the probability of a favorable survey result?**
- **That is, if the probability of a favorable result (*p* = 0.45) where to change, would we make the same decision?**
- **How much could it change before we would make a different decision?**

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Sensitivity Analysis

p = probability of a favorable survey result

$(1 - p)$ = probability of a unfavorable survey result

$$\text{EMV(node 1)} = (\$106{,}400)p + (\$2{,}400)(1 - p)$$
$$= \$104{,}000p + \$2{,}400$$

We are indifferent when the EMV of node 1 is the same as the EMV of not conducting the survey, \$40,000

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$$\$104{,}000p + \$2{,}400 = \$40{,}000$$
$$\$104{,}000p = \$37{,}600$$
$$p = \$37{,}600/\$104{,}000 = 0.36$$

p >0.36, the decision will stay the same

p< 0.36, do not conduct survey

Decision Making With Experimentation Another Example

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Decision Problem with Experimentation: An Example

A Company has two decision alternatives:
1) Restructure Business and 2) Carry on Normal Business

There are two States of Nature in Business Environment:
1) Smooth No Risk and 2) Troublesome High Risk

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	State of Nature	
Alternatives	Smooth (S) No Risk	Troublesome (T) High Risk
Restructure Business (R)	800	-100
Normal Business (N)	150	100
Prior Probability	0.25	0.75

Decision Making with Experimentation

- Consider the Decision Tree Problem on Company Decision with Appointing a Consultant.
- The Consultant Opinion is an experimentation.
- The company now has two decisions to make:

  1) Whether to go for the Consultant

  2) If Consultant Opinion is assertive, then what company decision Else what decision?

- Here the second decision dependent upon the outcome of the first.

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Decision Making with Experimentation

	State of Nature	
Alternatives	Smooth (S) No Risk	Troublesome (T) High Risk
Restructure Business (R)	800	-100
Normal Business (N) (As Usual)	150	100
Prior Probability	0.25	0.75

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- Before arriving at the company decision, the company has option to seek a consultant opinion at a cost of 50.
- Information from consultant could help in deciding which alternative to pursue (restructure or Normal Business)

Prior and Conditional Probabilities

- **Seeking Consultant Opinion is the experimentation conducted at a cost of 50.**
- **However, the Consultant will not be able to predict the state of nature with 100% accuracy.**
- **Consultant can only give additional information on the probable states of nature in the form of the following conditional probabilities:**

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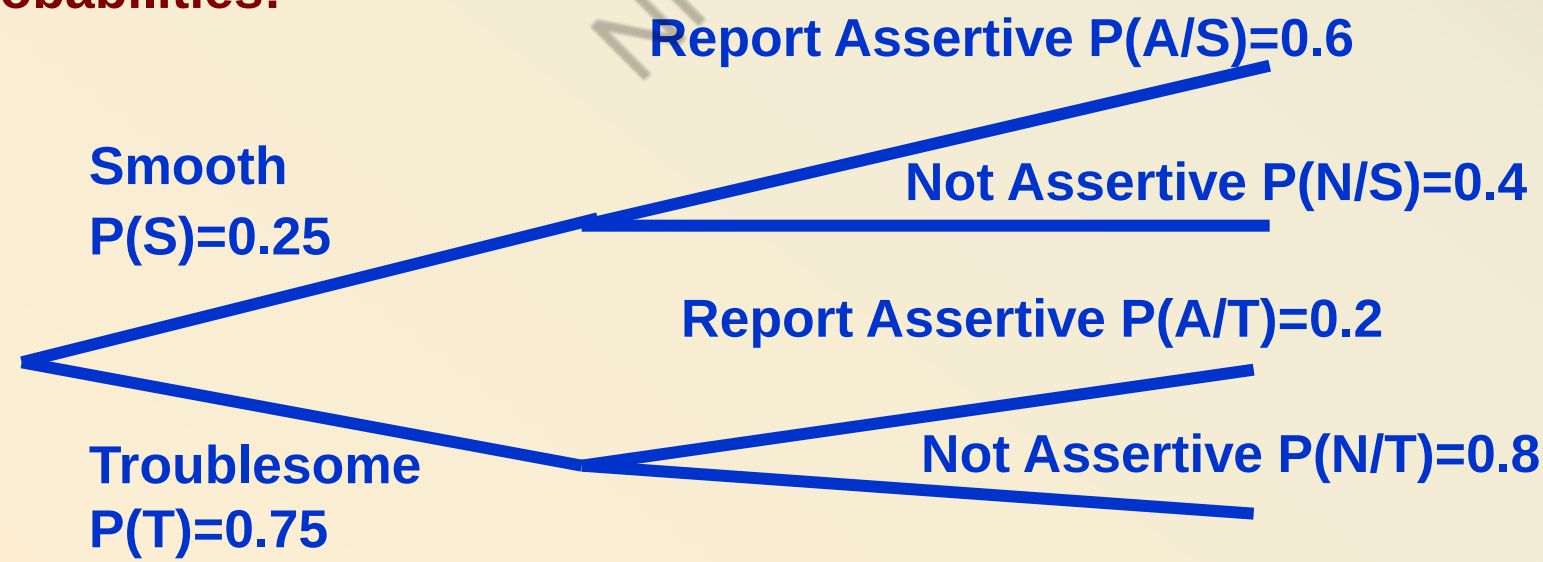
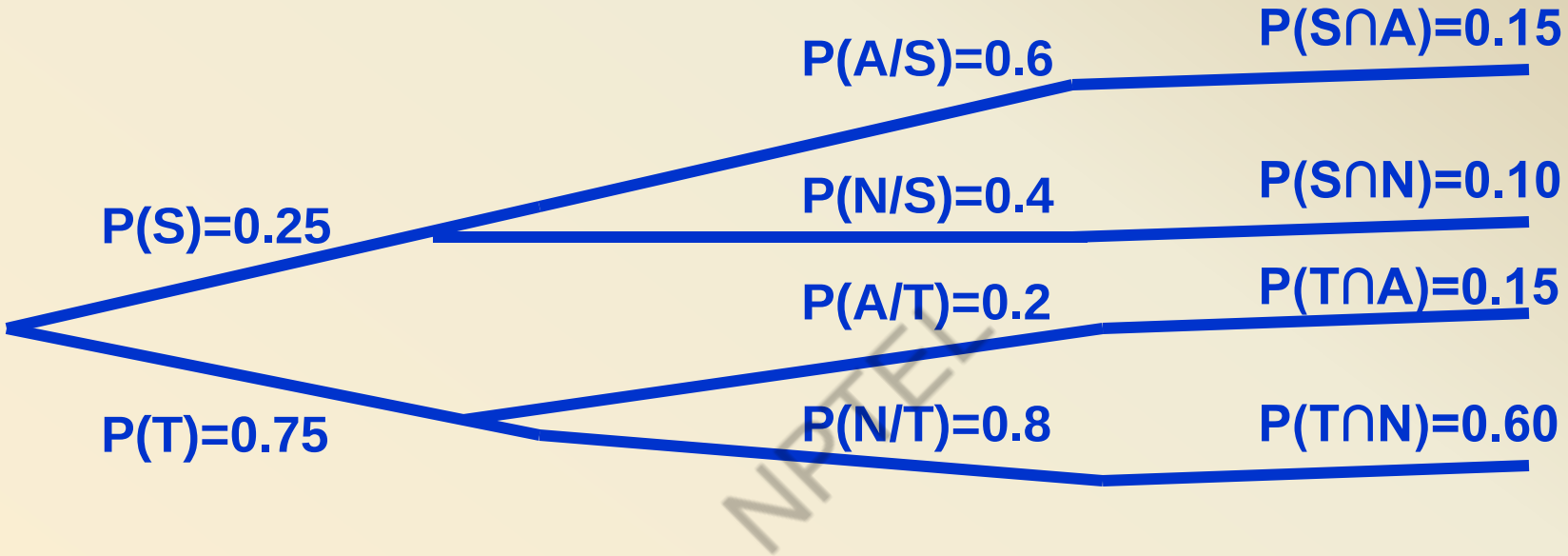


Baysean Analysis



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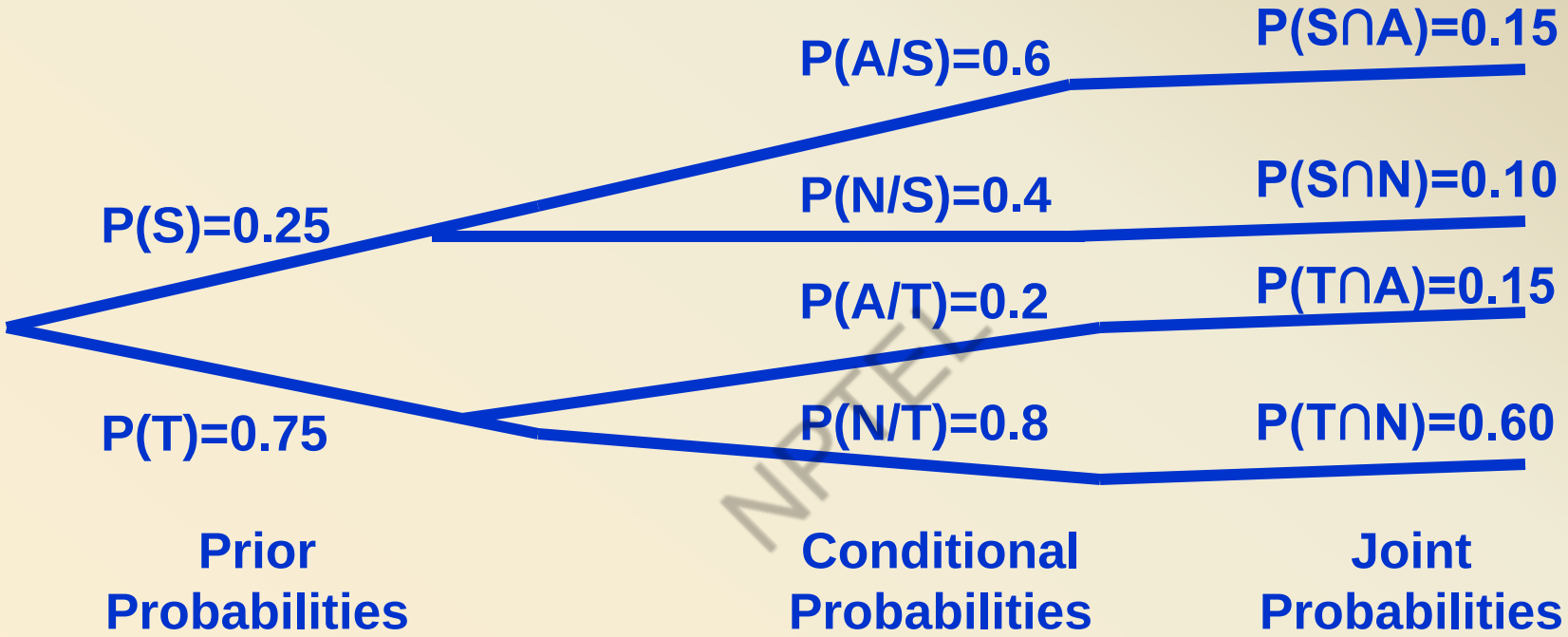
Prior Probabilities	Conditional Probabilities	Joint Probabilities

From Joint Probabilities, we obtain Unconditional Probabilities:

P(Assertive) = P(A) = P(S∩A) + P(T∩A) = 0.15 +0.15 = 0.30

P(Not Asserted) = P(N) = P(S∩N) + P(T∩N) = 0.10 +0.60 = 0.70

Baysean Analysis



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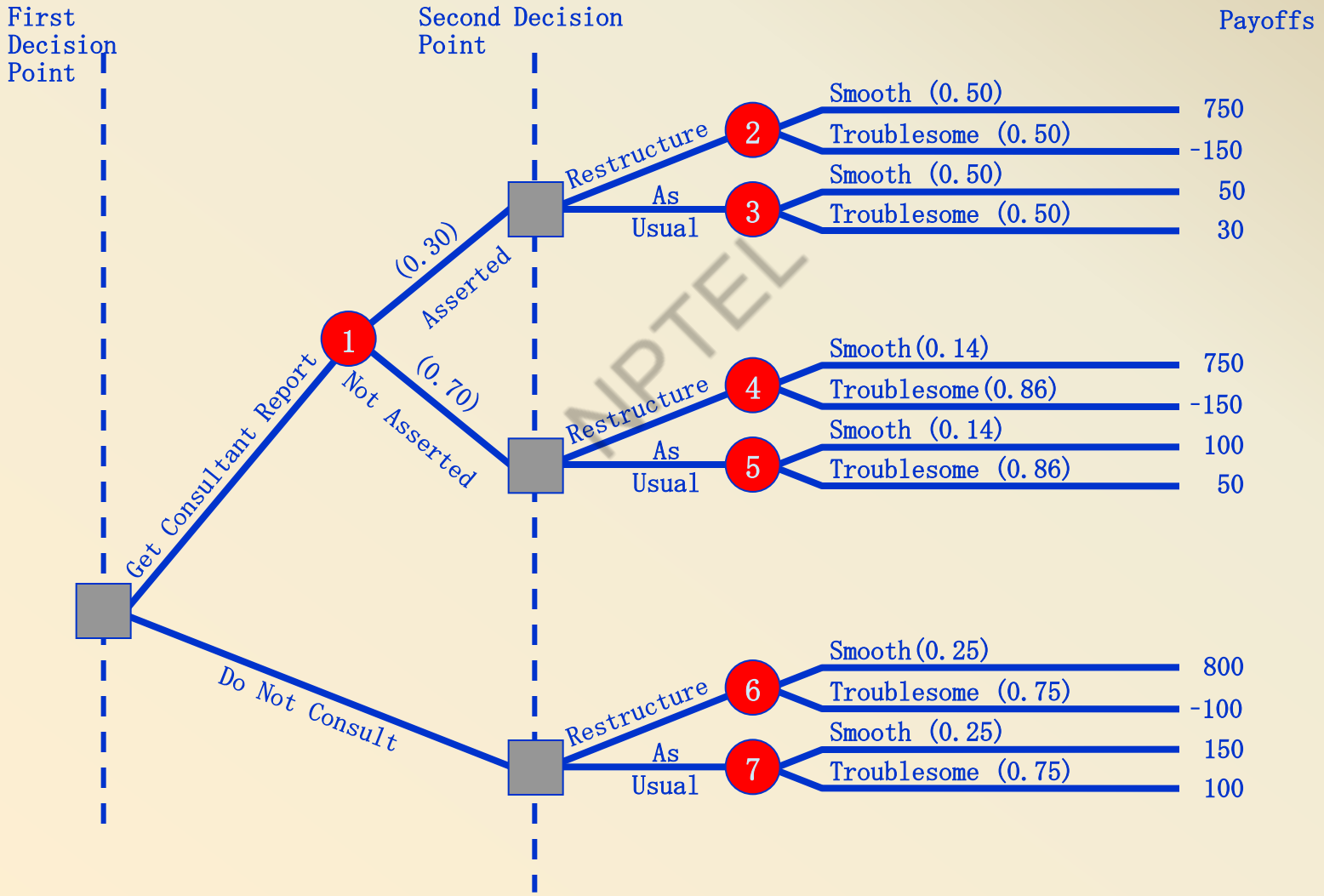
Unconditional Probabilities: P(A) = 0.30; P(N) = 0.70

Posterior Probabilities:

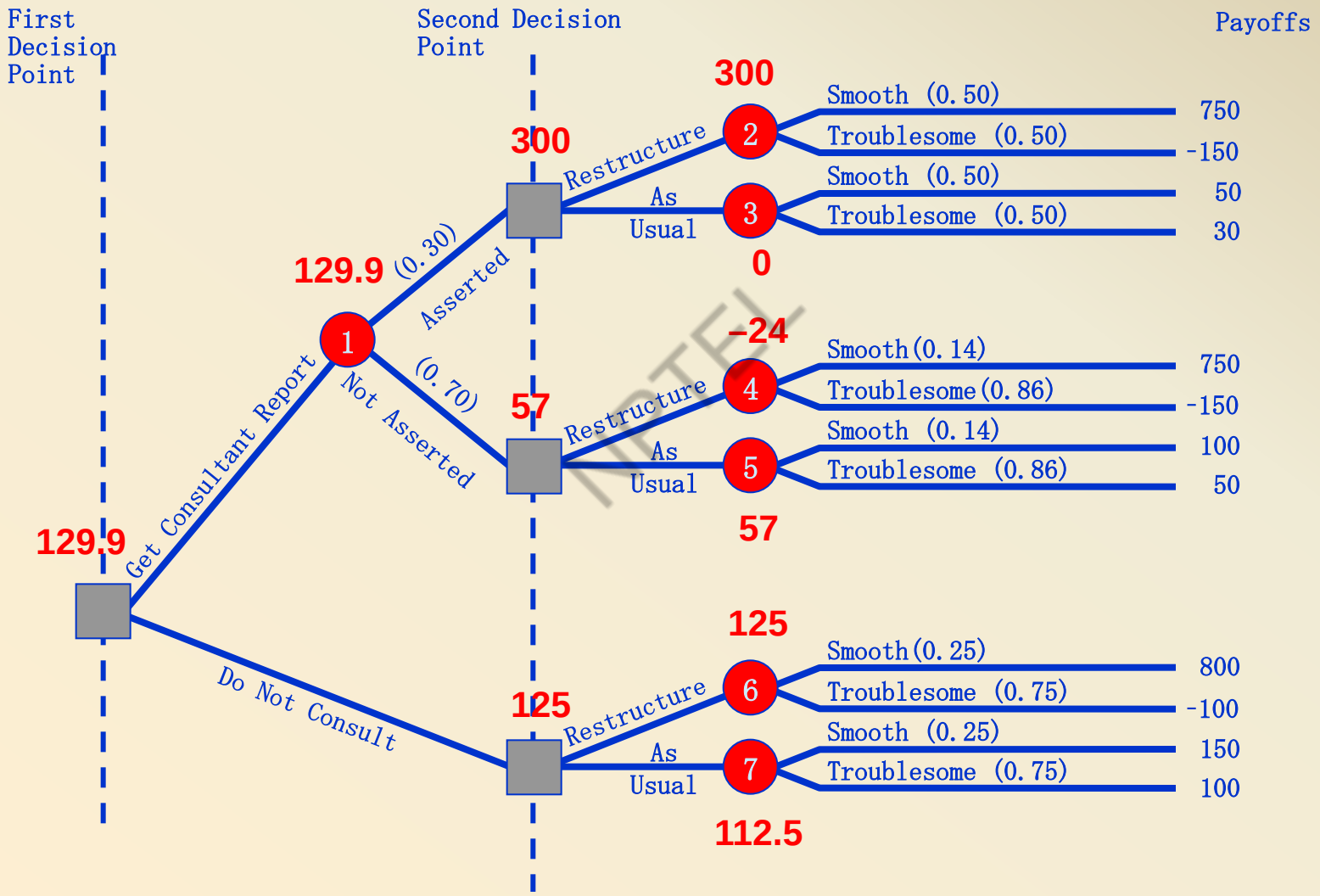
P(S/A)=P(S∩A)/P(A)=0.15/0.30=0.50; P(T/A)=P(T∩A)/P(A)=0.15/0.30=0.50

P(S/N)=P(S∩N)/P(N)=0.10/0.70=0.14; P(T/N)=P(T∩N)/P(N)=0.60/0.70=0.86

Revised Decision Tree

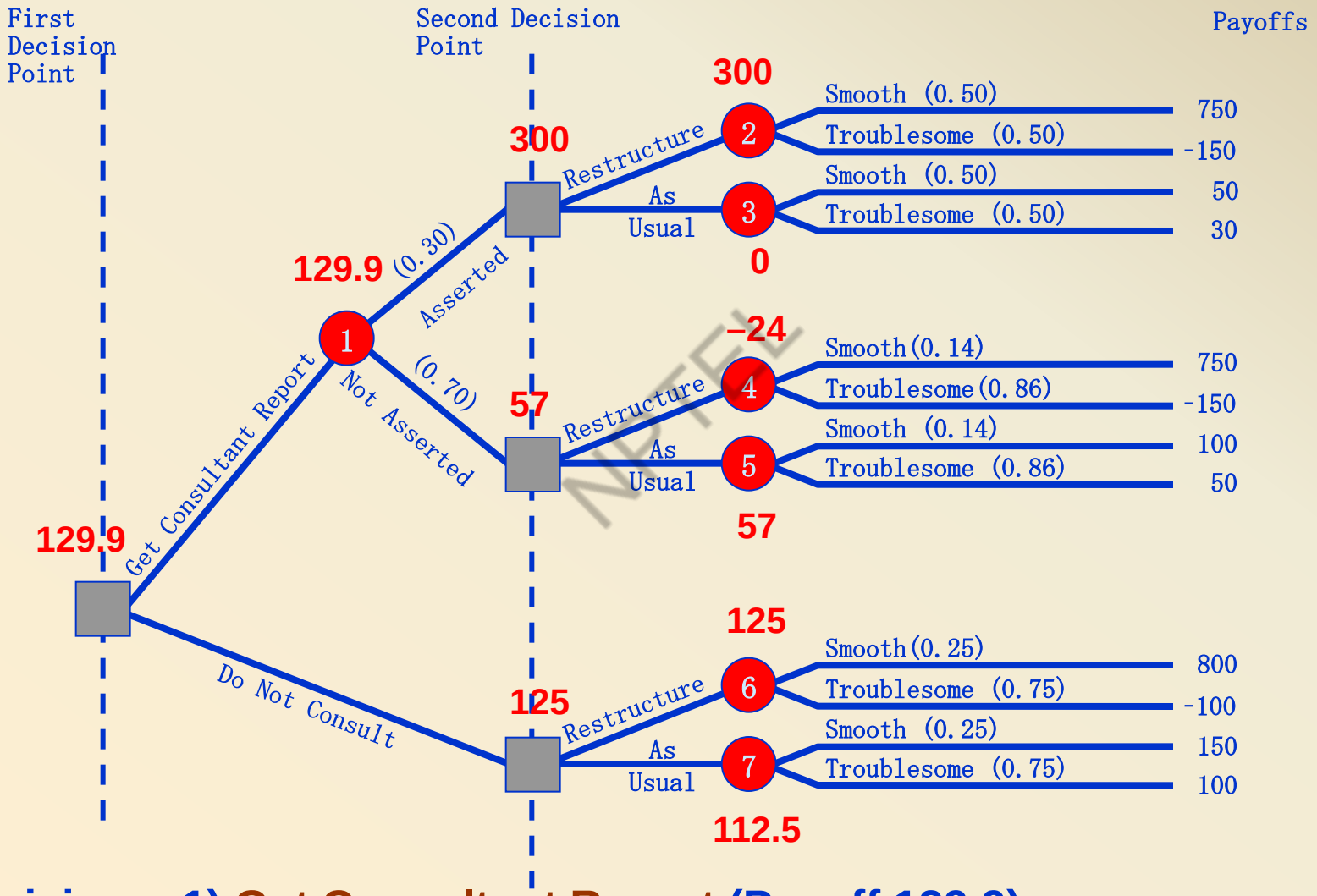


Payoffs and Decisions



What are the Decisions?

Payoffs and Decisions



Decisions: 1) Get Consultant Report (Payoff 129.9)
2) If Asserted then Restructure (Payoff 300); Else As Usual (Payoff 57)

THANK YOU

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